

SHRUBS

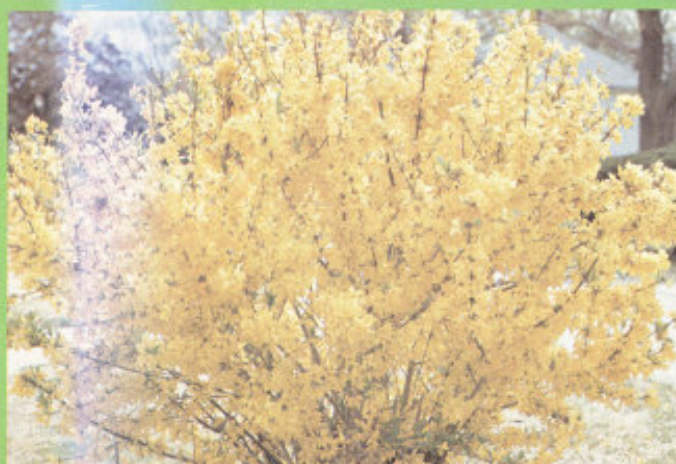
for KANSAS LANDSCAPES



Cooperative Extension Service ■ Kansas State University ■ Manhattan

SHRUBS

for KANSAS LANDSCAPES



Cooperative Extension Service ■ Kansas State University ■ Manhattan

The Project Gutenberg eBook of Shrubs for Kansas Landscapes

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Shrubs for Kansas Landscapes

Author: Gary L. McDaniel

Release date: May 26, 2020 [eBook #62234]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/62234

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK SHRUBS FOR
KANSAS LANDSCAPES ***

SHRUBS

for KANSAS LANDSCAPES

**Cooperative Extension Service • Kansas State University •
Manhattan**

This publication is one of a series of circulars to help you with your landscape planning. Use this bulletin when selecting shrubs for particular locations in your yard. Consider each plant for its intended use, site location, and exposure. Use the plant descriptions listed in the tables while visiting nurseries. This will help you to realize the size and performance you may expect from the plants. 2

All shrubs that will grow in the various sections of Kansas are not listed. Most of the shrubs mentioned were selected from the various Kansas retail nursery catalogs. A few shrub varieties are not listed in these retail catalogs, but your nurseryman can get them from wholesale nurseries supplying Kansas firms. The varieties listed are the more common shrubs used for landscaping purposes and will provide you with a wide assortment of plant material to solve any landscaping need.

GARY L. McDANIEL
Extension Specialist
Landscaping and Environmental Horticulture

The author wishes to thank Larry D. Leuthold for helpful comments during the preparation of this bulletin and Joe D. Roller for designing the illustrations used in this publication.



USES FOR WHICH WE WANT PLANT MATERIAL

1. Accent
2. Border
3. Soften Building Appearance
4. Screen poor views
5. Privacy

6. Hold a Soil Bank
7. ground cover

Shrubs are used in landscaping to provide low foliage and flowers that will compliment the house and trees in your yard. Each shrub used in your yard should fulfill a specific purpose for the overall design.

Select shrubs that will provide *accent* for your entryway or border focal point, *soften* corners and harsh lines, or provide a low *transition* from one point to another. Shrubs are also useful for dividing spaces by *separating* use areas in the yard. Shrubs are used with trees to *screen* unpleasant views or protect our yards from strong winds.

Shrubs as

Accent Plants

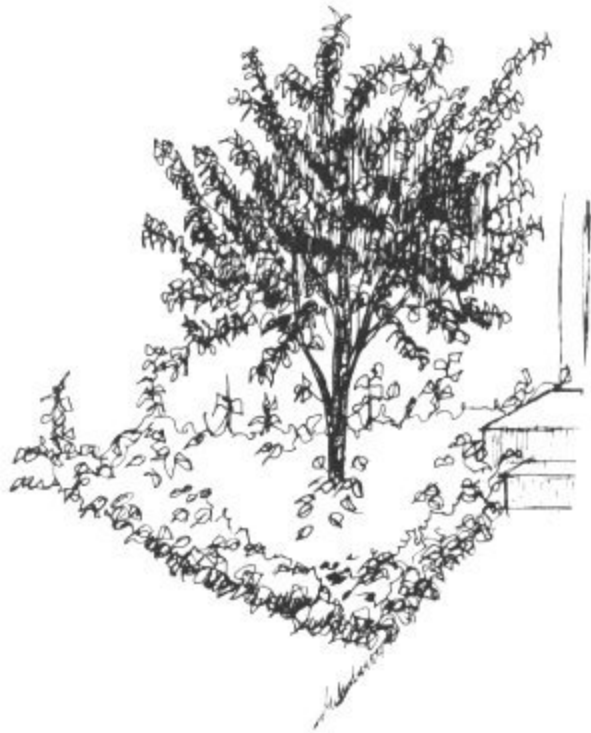


Accent end of walk if sidewalk approaches from driveway.

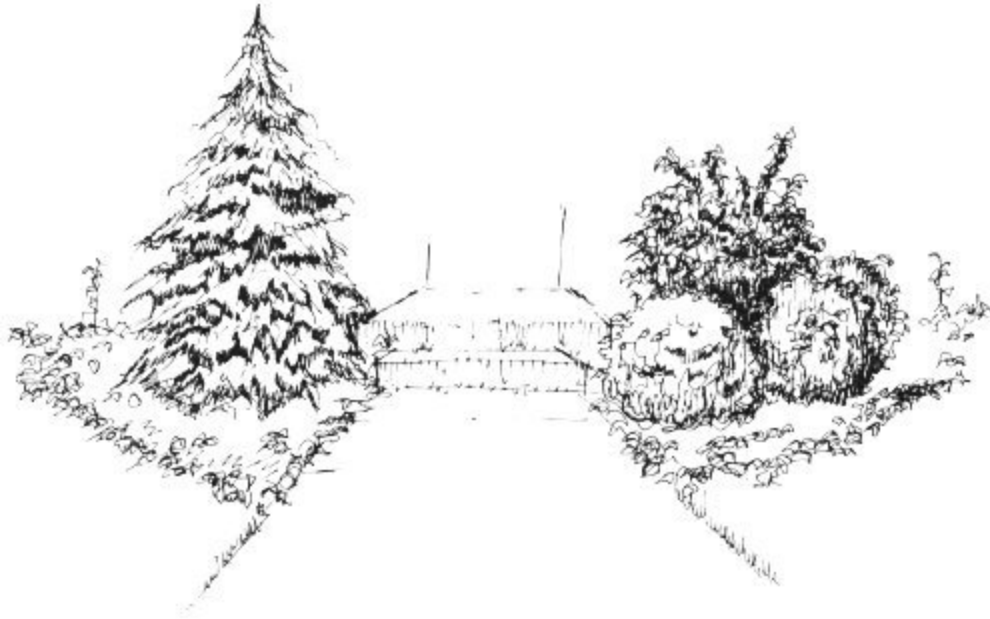
Shrubs that create a definite accent must be used with more care in your landscape than shrubs that soften lines. Two primary areas in your landscape should be accented. The most important of these is the front entryway to your home.

If the sidewalk to your front door approaches from the driveway along the front of the house, the accent area is at the end of the walk. This will provide a “visual” as well as a “physical” barrier that will point out the front door. Sidewalks that lead directly from the street to the door present a different situation. Accent may be created on either side of the front door, but never make the plantings completely symmetrical on each side. Create a strong accent to one side and a less dominant accent on the other side of the entryway. Never use accent plants on the corners of your house.

Accents may be created by (1) *Height*—taller plants catch the eye first; (2) *Color*—brightly colored foliage always attracts attention; (3) *Texture*—coarse leaf or stem textures provide strong contrasts; and (4) *Form*—pyramidal shrubs or clipping shrubs into unnatural shapes will create accent.

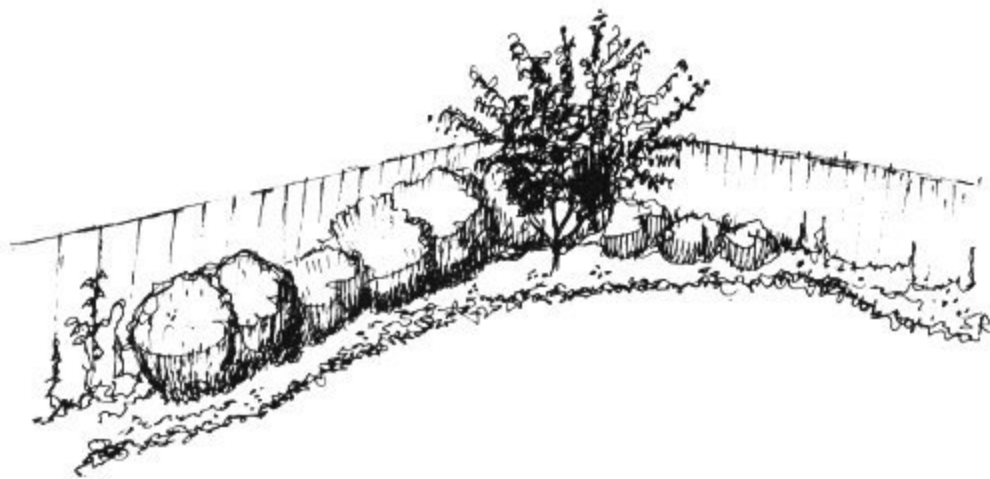


Shrubbery should create a stronger accent on one side of the entryway.



A single small upright Juniper can be used to accent an entry.

4



Another accent area for your landscape may be located in your shrub border. This area could be to the rear of your property, where it can be viewed from the patio or living room window. Never place accent plants randomly through the shrub border. This causes each accent plant to “fight” for dominance.

ACCENT PLANTS

DECIDUOUS SHRUBS: Tall (10 feet or more in height).

Elder, Golden
Fringe Tree, Red
Privet, Golden
Wayfaring Tree

DECIDUOUS SHRUBS: Medium (between 7 and 10 feet in height).

Burning Bush, Winged
Dogwood, Coral
Dogwood, Redosier
Dogwood, Silky
Dogwood, Yellowtwig
Ninebark, Common
Ninebark, Goldleaf
Sand Cherry, Purpleleaf
Viburnum, Arrowwood
Viburnum, Doublefile
Weigela

DECIDUOUS SHRUBS: Small (between 4 and 6 feet in height).

Barberry, Redleaf
Burning Bush, Dwarf Winged
Weigela, Dwarf

DECIDUOUS SHRUBS: Dwarf (less than 4 feet in height).

Barberry, Crimson Pygmy
Ninebark, Dwarf

BROADLEAF EVERGREEN SHRUBS: Tall (10 feet or more in height).

Holly
Magnolia, Southern

BROADLEAF EVERGREEN SHRUBS: Medium (between 7 and 10 feet in height).

Mahonia, Oregon Grape
Viburnum, Leatherleaf

BROADLEAF EVERGREEN SHRUBS: Small (between 4 and 6 feet in height).

Euonymus, Emerald 'n' Gold
Holly
Nandina

BROADLEAF EVERGREEN SHRUBS: Dwarf (less than 4 feet in height).

Azalea

Euonymus, Emerald Gaiety

Holly, Dwarf

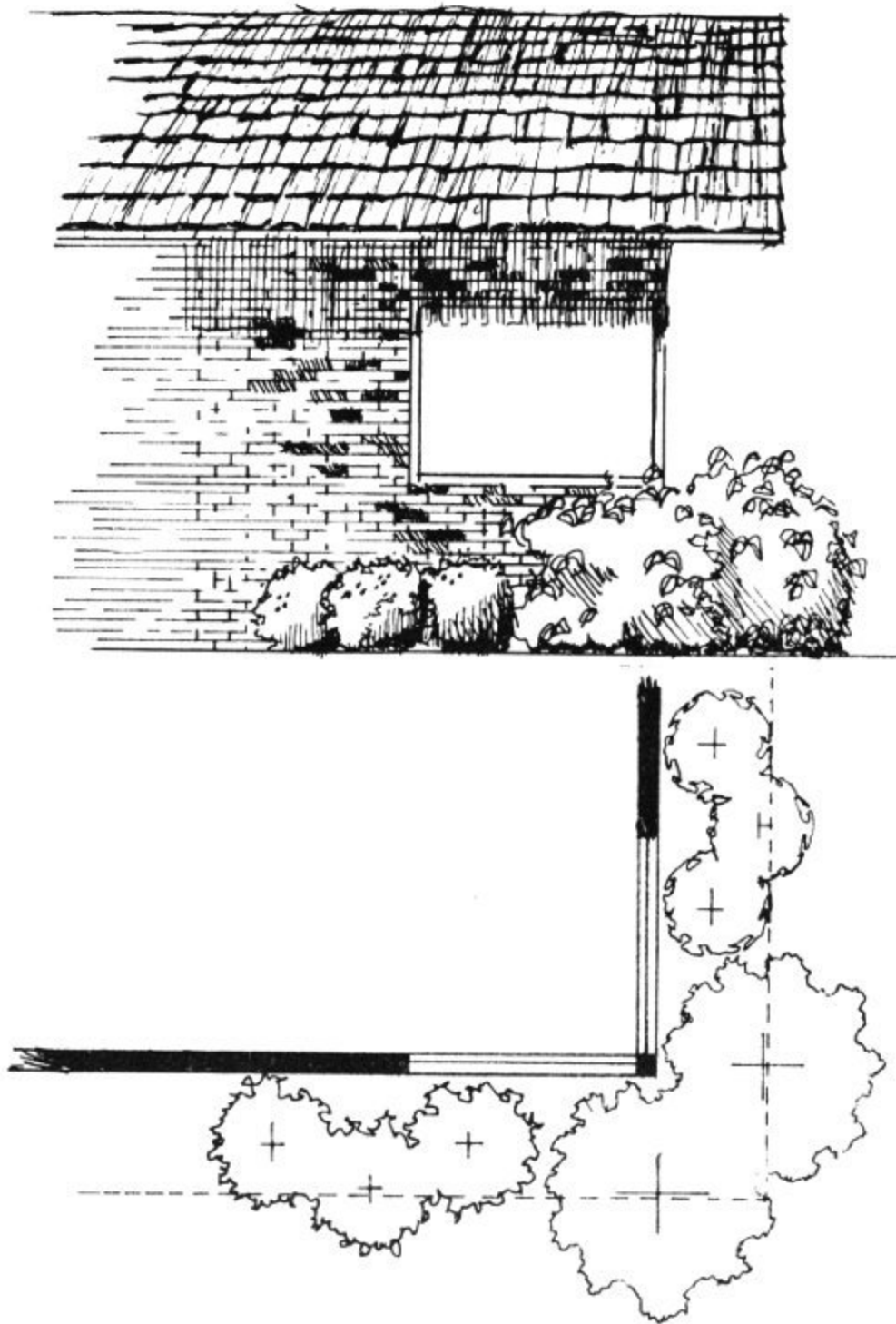
Mahonia, Compact Oregon Grape

Rhododendron

Shrubs that

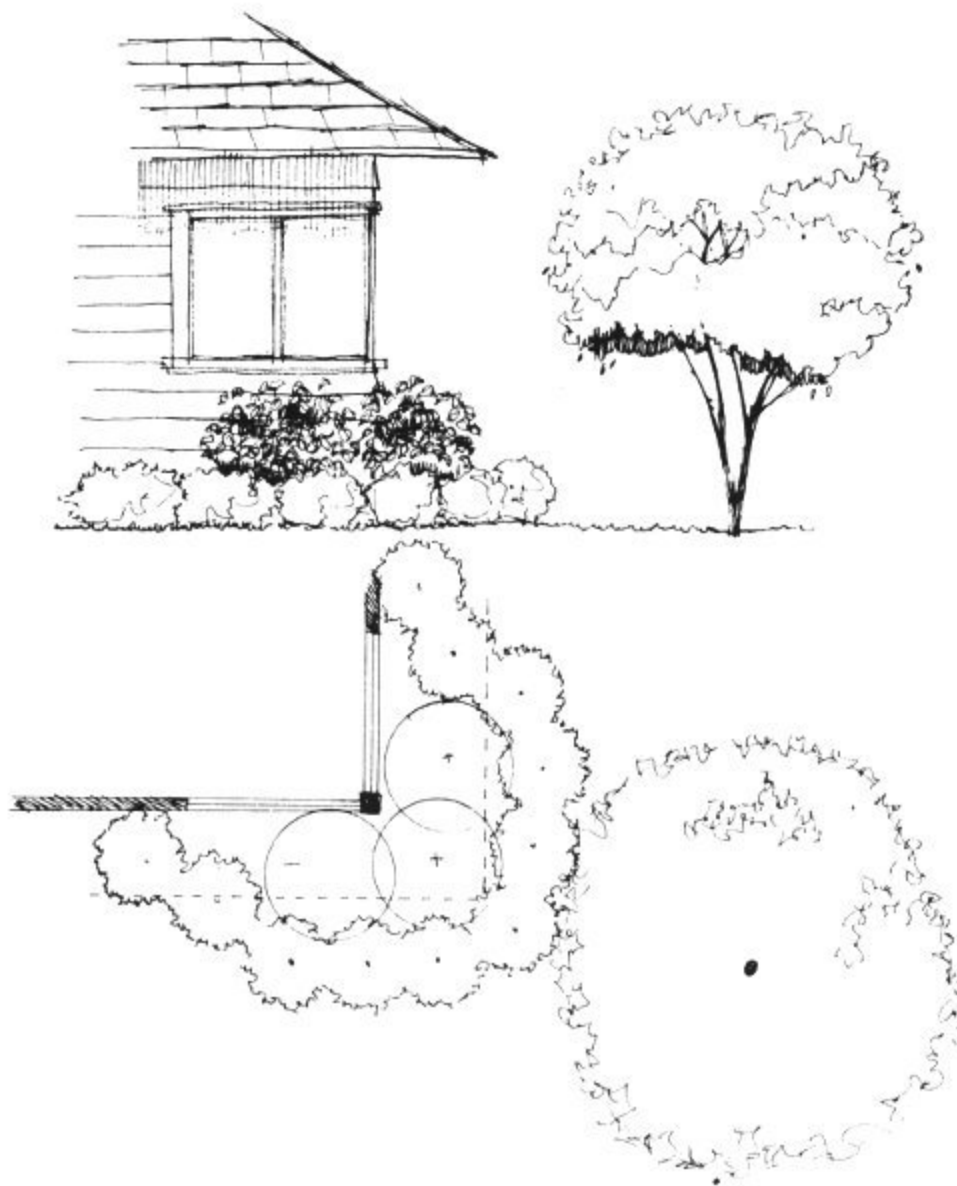
Soften the Building

Shrubs that will soften the lines of the house or buildings have a *round*, *oval*, or *irregular* shape. These shrubs break the strong vertical lines at the corners of the house, rather than call attention to them.



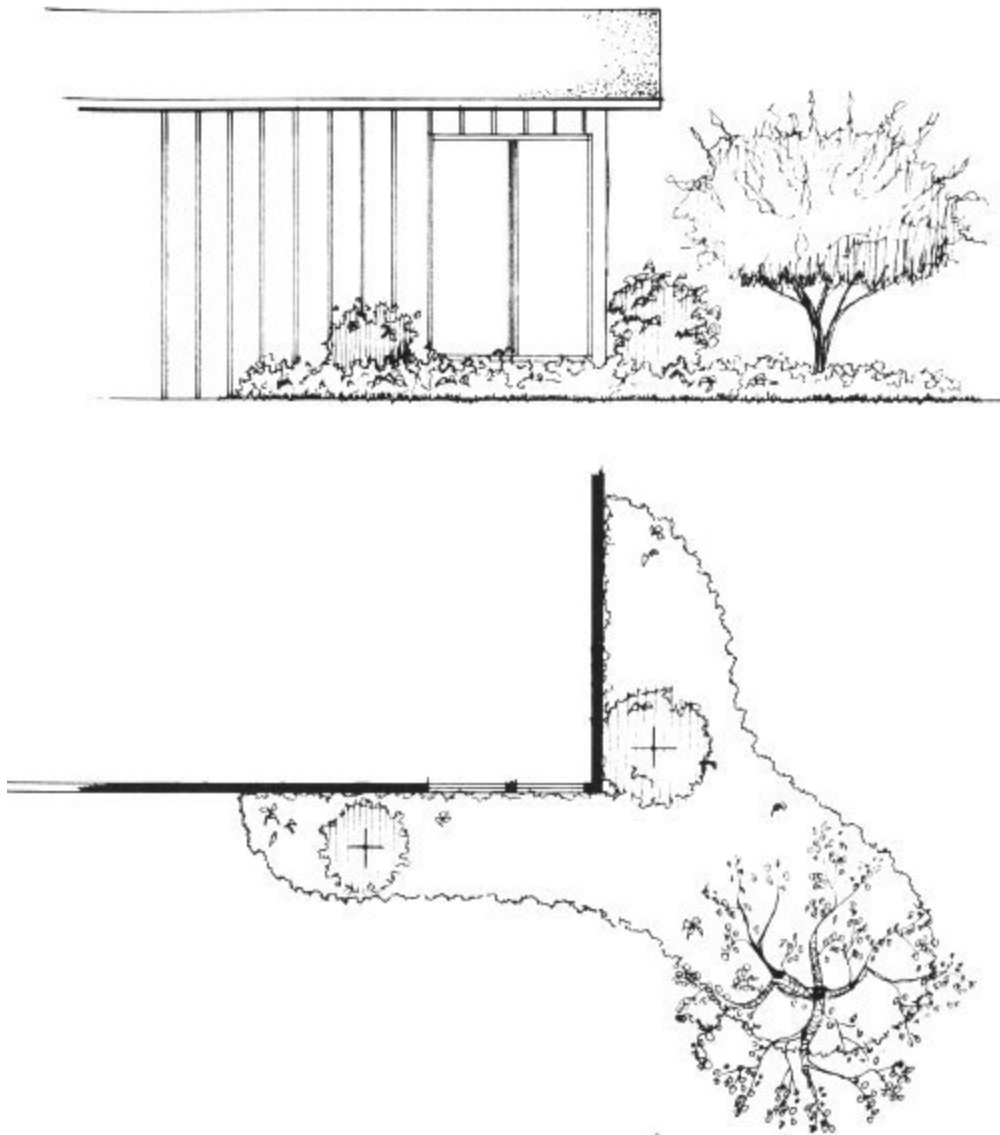
Shrubs will soften the lines at the corners of your house. Where windows are close to the corner, select dwarf shrubs that will not grow higher than the sill.

Never select shrubs that create an accent for corner plantings. Consider a brightly colored foliage on shrubs during the summer or fall as an accent. Do not plant upright-growing pyramidal evergreens, such as Junipers, for planting at the corners of your home. The shape of these plants creates an accent which will detract from the foundation planting. Instead of softening the harsh vertical lines at the corners of the house, pyramidal-shaped shrubs call attention to them. [5]



Use small ornamental trees to add a softer appearance to wall corners. Place these trees far enough away from the house that the

branches will not rub against it when the tree is fully grown. Place low facer plants in front of the taller shrubs to give the corner planting more depth.

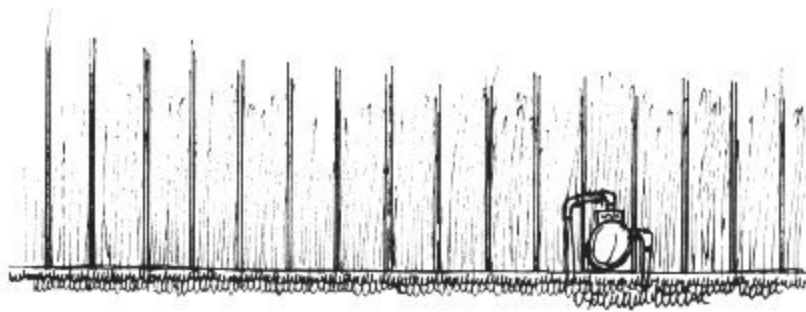


Floor length windows at the corners require low groundcovers or very dwarf shrubs below them. A small ornamental tree will assist in softening the corner, if you want some shade on the window. Extending the groundcover around the tree will make the mowing much easier.

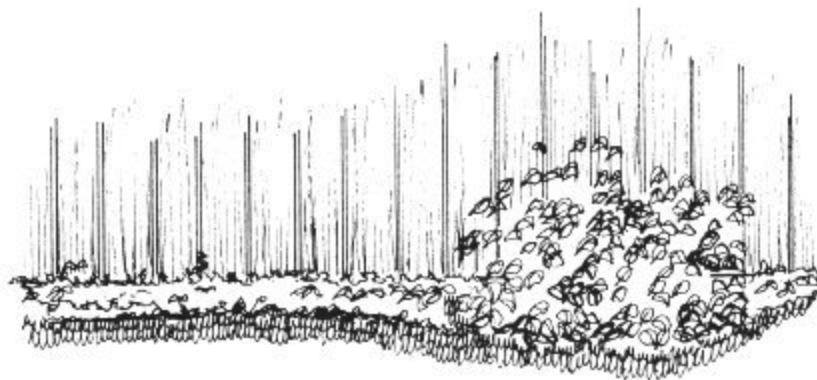
TRANSITION PLANTINGS

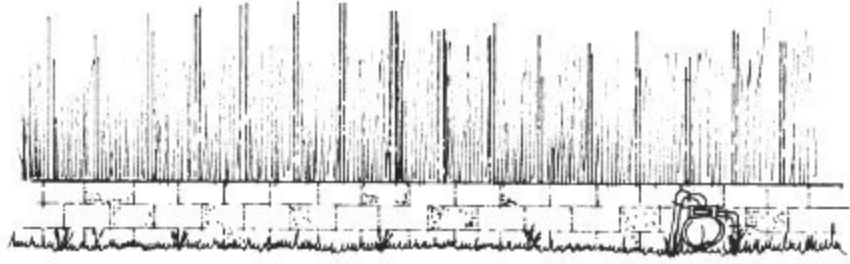
Plants used as a transition are simply those shrubs or ground covers that pull together the plantings at the corners or accented entryways. These shrubs are used to soften the lines of the house, where the walls meet the ground.

Use shrubs that will not call attention to themselves by having bright foliage or coarse leaf texture. These shrubs also help to hide any exposed foundation walls along the house. Transition shrubs should be low-growing or dwarf in size. If the siding of your house extends to the ground without leaving the foundation walls exposed, a groundcover may be all that is required to add depth to the foundation planting.

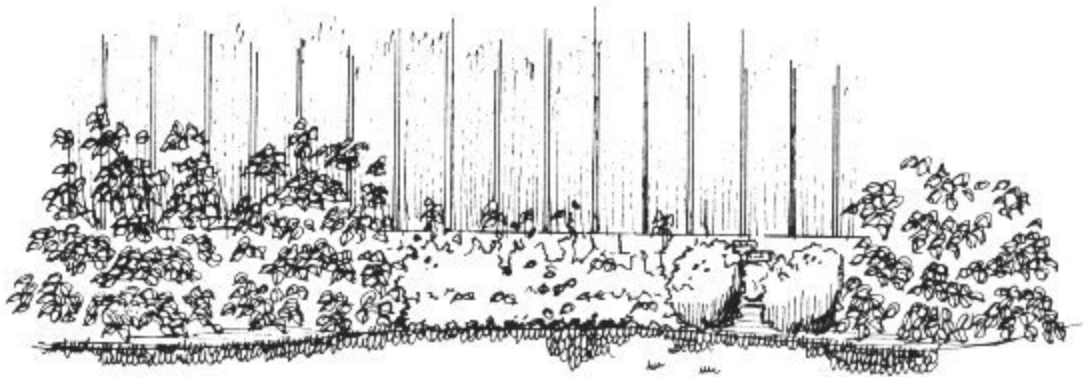


Shrubs help hide unsightly construction





... and soften harsh house lines.



CONTINUOUS BLOOM FOR YOUR LANDSCAPE

Whether you are selecting shrubs for the foundation plantings around the house or for a shrub border in your yard, choose shrubs that will bloom at different seasons. Most of the flowering shrubs that are available bloom during the spring. Once some shrubs have finished blooming, they have no other ornamental features. The interest in your landscape will be lost, unless you have selected some summer-blooming shrubs. Attempt to design your landscape so you can have continuous shrub bloom from early spring through late summer.

7

SHRUB ORDER OF BLOOM

EARLY SPRING

DECIDUOUS SHRUBS (Medium)

Redbud Chinese

Willow, Pussy

DECIDUOUS SHRUBS (Tall)

Cherry, Nanking

Crabapple, Sargent

Forsythia

Magnolia, Star

DECIDUOUS SHRUBS (Small)

Honeysuckle, Winter (Fragrant)

Quince, Flowering

DECIDUOUS SHRUBS (Dwarf)

Quince, Japanese Flowering

Forsythia, Bronx

BROADLEAF EVERGREEN SHRUBS

Magnolia, Southern

LATE SPRING

DECIDUOUS SHRUBS (Tall)

Cranberrybush

Dogwood, Gray

Honeysuckle, Tatarian

Lilac

Maple, Amur

Olive, Autumn

Pearlbush

Pea Shrub

Plum, Flowering

Snowball

Viburnum

Wayfaring Tree

DECIDUOUS SHRUBS (Medium)

Barberry

Bayberry, Northern

Beautybush

Cotoneaster, Large Flowering
Deutzia, Lemoine
Dogwood
Honeysuckle, Morrow
Honeysuckle, Zabel
Jetbead, Black
Privet
Sand Cherry, Purpleleaf
Scotch Broom
Spirea, Bridalwreath
Spirea, Vanhoutte
Viburnum

DECIDUOUS SHRUBS (Small)

Almond, Flowering
Barberry, Japanese
Cotoneaster, Spreading
Kerria
Lilac, Dwarf Korean
Spirea, Garland
Spirea, Thunberg
Weigela

DECIDUOUS SHRUBS (Dwarf)

Barberry, Crimson Pygmy
Cotoneaster, Cranberry
Deutzia
Pea Shrub, Pygmy
Quince, Japanese Flowering
Spirea, Snowmound

BROADLEAF EVERGREEN SHRUBS

Barberry, Wintergreen
Mahonia, Oregon Grape
Pyracantha
Viburnum, Leatherleaf

EARLY SUMMER

DECIDUOUS SHRUBS (Tall)

Elder, Golden
Fringe Tree, Red
Mockorange, Big Scentless
Privet
Smoke Tree (Purple Fringe)
Tamarisk

DECIDUOUS SHRUBS (Medium)

Deutzia
Lilac, Late
Lilac, Persian
Mockorange
Viburnum

DECIDUOUS SHRUBS (Small)

Beautyberry, Japanese
Hydrangea, Annabelle
Mockorange
Spirea, Anthony Waterer
Spirea, Froebel
Spirea, Redflowering Japanese

BROADLEAF EVERGREEN SHRUBS

Euonymus (Patens)

LATE SUMMER

DECIDUOUS SHRUBS (Tall)

Crape myrtle
Rose-of-Sharon
Tamarisk

DECIDUOUS SHRUBS (Medium)

Butterfly Bush
Tamarisk, 'Summer Glow'

DECIDUOUS SHRUBS (Small)

Abelia, Glossy
Hydrangea, A. G.
Mockorange

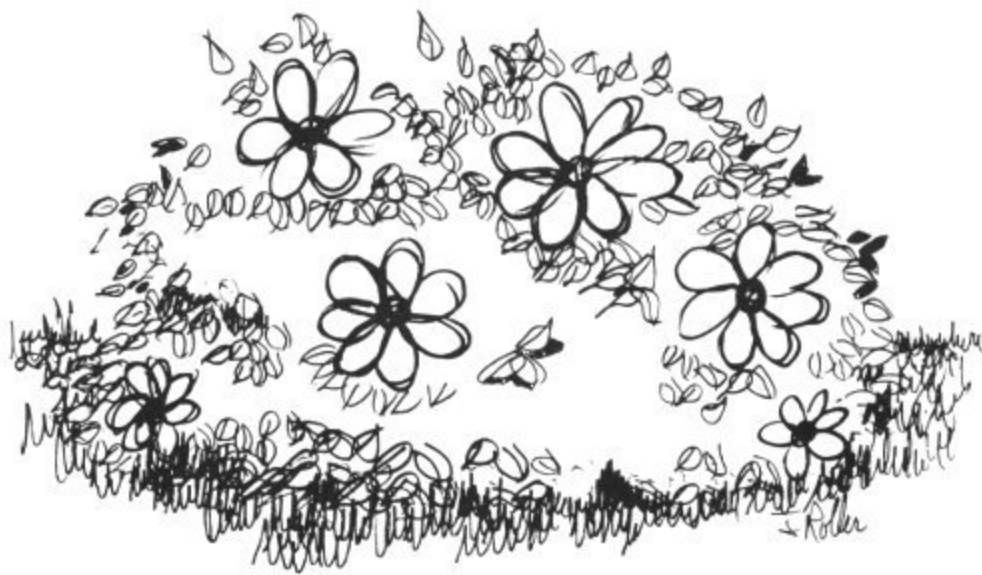
DECIDUOUS SHRUBS (Dwarf)

Chenault Coralberry

Cinquefoil

Hypericum

Spirea, Anthony Waterer



8

SHRUB BLOOM COLOR

DECIDUOUS SHRUBS (Tall)

WHITE

Cranberrybush

Crapemyrtle

Dogwood, Gray

Elder, Golden

Honeysuckle, White Tatarian

Lilac, Common

Lilac, Peking

Pearlbush

Privet
Redbud, Chinese
Rose-of-Sharon
Snowball
Viburnum

PINK

Crapemyrtle
Fringe Tree, Red
Honeysuckle, Red Tatarian
Honeysuckle, Rosy Tatarian
Plum, Flowering
Rose-of-Sharon
Smoke Tree (Purple Fringe)
Tamarisk
Willow, Pussy

YELLOW

Barberry, Mentor
Maple, Amur
Olive, Autumn
Pea Shrub

PURPLE-LILAC

Lilac, Chinese
Lilac, Common
Rose-of-Sharon

DECIDUOUS SHRUBS (Medium)

WHITE

Butterflybush
Crabapple, Sargent
Deutzia, Lemoine
Deutzia, Pride of Rochester
Dogwood
Honeysuckle, Morrow
Jetbead, Black
Magnolia, Star

Mockorange
Privet
Sand Cherry, Purpleleaf
Scotch Broom
Spirea, Bridalwreath
Spirea, Vanhoutte
Viburnum

PINK

Bayberry, Northern
Beautybush
Butterfly Bush
Cherry, Nanking
Cotoneaster, Large Flowering
Crabapple, Sargent
Deutzia, Pride of Rochester
Honeysuckle, Zabel
Lilac, Late
Lilac, Persian
Magnolia, Pink Star
Scotch Broom
Tamarix, 'Summer Glow'
Weigela

YELLOW

Barberry
Forsythia
Scotch Broom

DECIDUOUS SHRUBS (Small)

WHITE

Almond, Flowering
Honeysuckle, Winter (Fragrant)
Hydrangea, A. G.
Mockorange
Quince, Japanese Flowering
Spirea, Garland

Spirea, Thunberg

Viburnum

PINK

Abelia, Glossy

Almond, Flowering

Beautyberry, Japanese

Cotoneaster, Spreading

Hydrangea

Quince, Japanese Flowering

Weigela

YELLOW

Barberry, Japanese

Kerria

PURPLE

Hydrangea (blue)

Lilac, Dwarf Korean

DECIDUOUS SHRUBS (Dwarf)

WHITE

Deutzia, Slender

Hydrangea, Annabelle

Mockorange, Dwarf

Spirea, Snowmound

PINK

Chenault Coralberry

Cotoneaster, Cranberry

Deutzia, Rosepanicle

Quince, Japanese Flowering (red)

Spirea, Froebel (crimson)

Spirea, Anthony Waterer

Spirea, Redflowering, Japanese

YELLOW

Barberry, Crimson Pygmy

Cinquefoil

Forsythia, Bronx

Hypericum
Pea Shrub, Pygmy
ORANGE
Quince, Japanese Flowering

BROADLEAF EVERGREEN SHRUBS

WHITE
Azalea
Euonymus (Patens)
Magnolia, Southern
Pyracantha
Rhododendron
Viburnum, Leatherleaf

LILAC
Rhododendron

PINK
Azalea
Rhododendron

YELLOW
Azalea
Barberry, Wintergreen
Mahonia, Oregon Grape

Shrubs for

Shady Locations

Most of the plants that are adapted for western Kansas will grow on the north side if they are planted 3 to 4 feet from the building and out from the roof overhang. In eastern Kansas it is necessary to consider shade tolerance when planting on the north side of the house.

Most shrubs will bloom more profusely and produce more fruit when planted in the sun, even though they may survive quite well in deep shade. Often shrubs will grow better in shade simply because they are protected from the southwest winds. If you don't want to plant dwarf shrubs 3 to 4 feet out from the foundation of the house, then plant only shade tolerant shrubs.

What is often thought of as a shade problem is often the result of competition from the roots of adjoining trees. Apply plant food and water and thin the tree branches to correct this problem.

The shrubs listed for shady locations may be expected to perform adequately on the north side of the house or behind fences in deep shade. Many of these same shrubs will do equally well in strong sunlight or shade.

DECIDUOUS SHRUBS: Tall (10 feet or more in height).

Mockorange, Big Scentless

Privet, Amur River North

Privet, California

Privet, Common

Viburnum, Cranberrybush

Viburnum, Snowball

Wahoo

DECIDUOUS SHRUBS: Medium (between 7 and 10 feet in height).

Barberry, Mentor
Barberry, Truehedge
Dogwood, Coral
Dogwood, Redozier
Dogwood, Silky
Dogwood, Yellowtwig
Holly, Winterberry
Jetbead, Black
Mockorange, Enchantment
Mockorange, Innocence
Mockorange, Minnesota Snowflake
Mockorange, Sweet
Ninebark
Spirea, Vanhoutte

DECIDUOUS SHRUBS: Small (between 4 and 6 feet in height).

Abelia, Glossy
Barberry, Japanese (green)
Barberry, Kobold
Honeysuckle, Winter (Fragrant)
Hydrangea, A. G.
Hydrangea, Nikko Blue
Indiancurrent Coralberry
Kerria, Japanese
Mockorange, Alberta and Glacier
Mockorange, Belle Etoile
Mockorange, Bouquet Blanc
Snowberry

DECIDUOUS SHRUBS: Dwarf (less than 4 feet in height).

Chenault Coralberry
Hydrangea, Annabelle
Mockorange, Dwarf Minnesota Snowflake
Mockorange, Silver Showers
Mockorange, Dwarf Sweet
Ninebark, Dwarf

BROADLEAF EVERGREEN SHRUBS: Tall (10 feet or more in height).

Holly, American
Holly, Yaupon

Magnolia, Southern

BROADLEAF EVERGREEN SHRUBS: Medium (between 7 and 10 feet in height).

Euonymus, Large Leaf

Euonymus, Manhattan

Holly, Burford

Mahonia, Oregon Grape

Viburnum, Leatherleaf

BROADLEAF EVERGREEN SHRUBS: Small (between 4 and 6 feet in height).

Euonymus, Dupont

Euonymus, Jewel

Euonymus, Medium Leaf

Euonymus, Sarcoxie

Holly, Dwarf Japanese

Holly, Heller's

Holly, Hetz

Nandina

BROADLEAF EVERGREEN SHRUBS: Dwarf (less than 4 feet in height).

Azalea, Kurume

Azalea, Mollis Hybrid

Boxwood, Edging

Boxwood, Wintergreen

Euonymus, Emerald Cushion

Euonymus, Emerald Gaiety

Euonymus, Small Leaf

Euonymus, Vinca Leaf

Holly, Dwarf Burford

Holly, Dwarf Chinese

Mahonia, Compact Oregon Grape

Rhododendron

SHRUBS HAVING FRUIT THAT WILL ATTRACT BIRDS

DECIDUOUS SHRUBS: Tall (10 feet or more in height).

Cotoneaster, Peking

Cranberrybush, American
Dogwood, Gray
Elder, Golden
Honeysuckle, Tatarian
Olive, Autumn
Privet, California
Privet, Common
Privet, Golden (Vicary)
Viburnum, Nannyberry
Wayfaring Tree

DECIDUOUS SHRUBS: Medium (between 7 and 10 feet in height).

Barberry, Truehedge
Cherry, Nanking
Dogwood, Coral
Honeysuckle, Morrow
Honeysuckle, Zabel
Jetbead, Black
Privet, Border
Privet, Regel
Sand Cherry, Purpleleaf
Viburnum, Arrowwood
Viburnum, Korean Spice
Viburnum, Linden

DECIDUOUS SHRUBS: Small (between 4 and 6 feet in height).

Barberry, Japanese
Beautyberry, Japanese
Cranberrybush, Dwarf American
Honeysuckle, Clavey Dwarf
Honeysuckle, Winter (Fragrant)
Viburnum, Wrights'

DECIDUOUS SHRUBS: Dwarf (less than 4 feet in height).

Chenault Coralberry
Cotoneaster, Cranberry

BROADLEAF EVERGREEN SHRUBS

Holly (female)
Pyracantha
Nandina

Low Maintenance

for your shrubs

HEIGHT AND SPREAD

Consider low maintenance when planning your landscape. When selecting shrubs for your yard, determine the maximum height and spread you want from the planting, and then select only from shrubs with those characteristics.

The shrubs listed in this bulletin have been grouped according to their mature heights. If you are looking for a shrub that will grow under a window that is 4 feet from the ground, select from those listed as dwarf shrubs. You certainly would not select a Tatarian Honeysuckle (10 to 12 feet) for this location.

Never select a shrub that will ultimately grow too high for its location and expect to keep it shorter by constant clipping. This will only add to your time in maintaining your landscape.

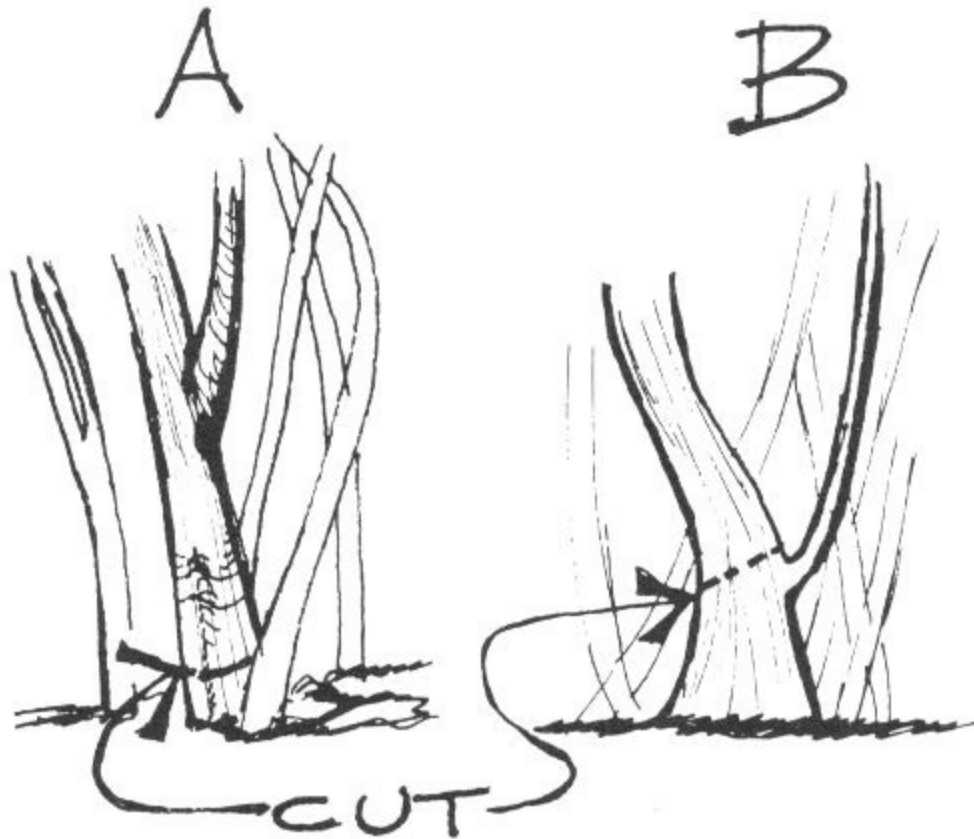


Wrong selection of shrubs can be disastrous as shown by the planting on the right.

MAINTAINING SHRUB SIZE

Shrubs will require periodic pruning to maintain their proper shape and size. Many shrubs will grow tall and spindly if not thinned-out occasionally. Some shrubs will sucker severely at the ground if the older stems or canes are not removed. This causes the shrub to spread beyond its normal limits.

The proper method for maintaining shrub shape is to remove all canes that are larger than one inch in diameter. These should be removed as close to the ground line as possible. This practice will open the center of the plant so that more light can penetrate into it. Also, the size of the shrub will be maintained at the desirable height.



When removal of a cane has been decided upon, make the cut as close to the ground as possible (example A). When an old cane selected for removal has a desirable shoot at its base, make a cut as shown in example B. Note that the cut is slanted and is approximately $\frac{1}{4}$ inch above the shoot.

ADAPTABILITY AND HARDINESS

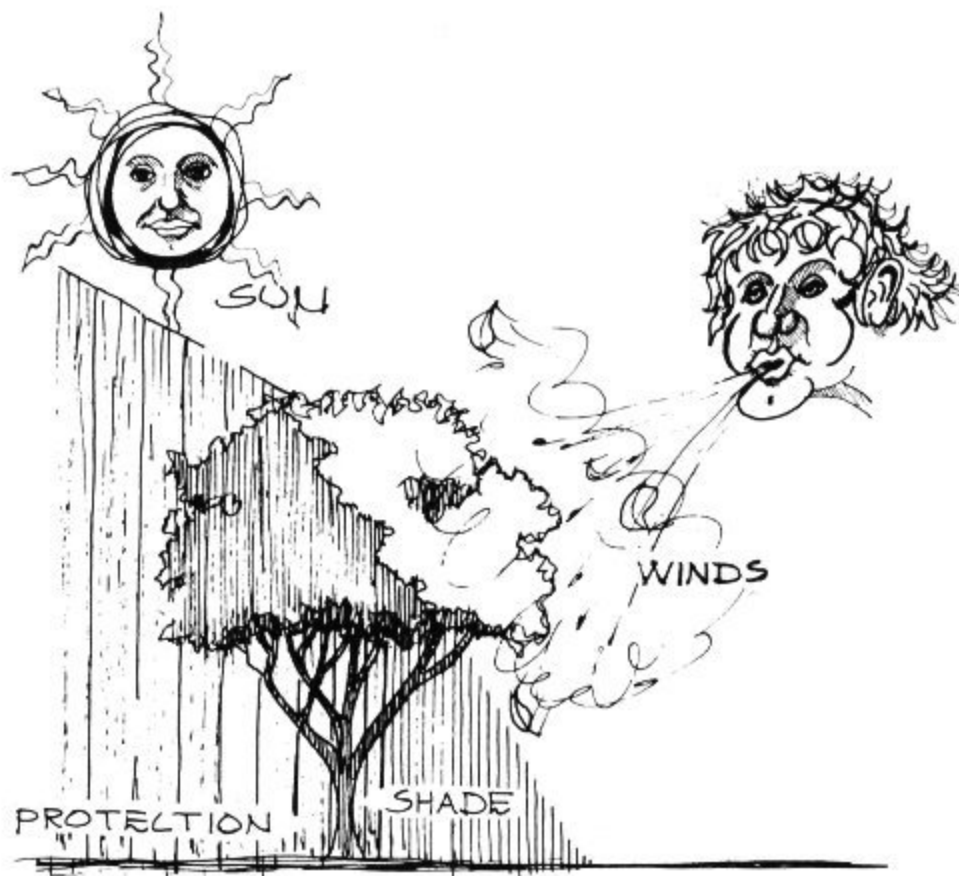
Which plants are best suited for your landscape? This will depend on whether the shrubs will be planted on a farmstead or in a city lot. It will also depend on whether your yard is exposed to direct southwest winds, or whether you have some protection.

Plant adaptability has been considered for each shrub listed in this bulletin. You may plant any shrub that is listed as 'Hardy' in an open, fully exposed location and expect it to survive. Those shrubs that have an 'X' will most likely not survive if planted where they will receive direct exposure to the hot winds of late summer. If you live in a town or city, you may have

enough wind protection to use these shrubs. This is most important for homeowners living in the western counties of Kansas. 11

Protection from southwest summer winds may be provided by a windbreak, trees in the city, by planting on the east or north side of the house, or behind a screen fence. If you have this protection you might be able to use plants that are not listed for your area. However, if your yard is exposed to hot summer winds, you had better use only the plant listed for your section of Kansas.

You will sometimes find plants growing in your area that are not listed for your particular section of Kansas. It is possible to grow some plants outside their recommended region, provided these plants receive adequate protection and care. These shrubs will usually not grow as large or as tall as they would further south or east. If you are rather certain that your yard will provide enough protection for these plants, you might try growing a few.



VARIETY LISTINGS OF SHRUBS

[1]

<i>Common Name</i>	<i>Botanical Name</i>	<i>Height</i>	<i>Spread</i>	<i>Adaptability</i>
DECIDUOUS SHRUBS: Tall (10 feet in height or more)				
Cotoneaster, Peking	<i>Cotoneaster acutifolia</i>	10-12'	8-10'	Hardy
Cranberrybush, American	<i>Viburnum trilobum</i>	12'	10-12'	Hardy
Cranberrybush, European	<i>Viburnum opulus</i>	8-12'	10-12'	Hardy
Crapemyrtle	<i>Lagerstroemia indica</i>	10-12'	5'	SE
Dogwood, Gray	<i>Cornus racemosa</i>	8-12'	8-12'	Hardy
Elder, Golden	<i>Sambucus canadensis 'Aurea'</i>	12'	10-12'	Hardy
Fringe Tree, Red	<i>Cotonis coggygria 'rubrifolia'</i>	10'	10-12'	X, P
Honeysuckle, Tatarian	<i>Lonicera tatarica</i>	10-12'	8-10'	Hardy
Hydrangea, P. G.	<i>Hydrangea paniculata 'grandiflora'</i>	10-15'	10-15'	X, P
Lilac, Chinese (Rothamagensis)	<i>Syringa chinensis</i>	10-15'	10-15'	Hardy
Lilac, Common	<i>Syringa vulgaris</i>	12-15'	12-15'	Hardy
Lilac, Manchurian	<i>Syringa amurensis</i>	12-15'	12-15'	Hardy
Lilac, Peking	<i>Syringa pekinensis</i>	10-15'	10-15'	Hardy
Maple, Amur	<i>Acer ginneis</i>	15-20'	10-12'	X, P
Mockorange, Big Scentless	<i>Philadelphus grandiflorus</i>	9-12'	5-8'	Hardy
Olive, Autumn	<i>Eleagnus umbellatus</i>	14-16'	15-17'	Hardy
Pea Shrub	<i>Caragana arborescens</i>	15-18'	12-15'	Hardy
Pearlbush	<i>Exochorda racemosa</i>	10-15'	10-15'	SC

Plum, Flowering	<i>Prunus triloba</i> 'plena'	12'	10-12'	X
Privet, Amur River North	<i>Ligustrum amurense</i>	10-12'	5-7'	Hardy
Privet, California	<i>Ligustrum ovalifolium</i>	10-12'	6-8'	X, P
Privet, Common	<i>Ligustrum vulgare</i>	10-12'	6-8'	Hardy
Privet, Golden (Vicary)	<i>Ligustrum X vicari</i>	12'	6-8'	X, P
Redbud, Chinese	<i>Cercis chinensis</i>	10-12'	10-12'	X, P
Rose-of-Sharon (Althea)	<i>Hibiscus syriacus</i>	8-12'	6-10'	Hardy
Smoke Tree (Purple Fringe)	<i>Cotinus coggygia</i>	15'	8'	X, P
Snowball	<i>Viburnum opulus</i> 'sterilis'	8-12'	10-12'	Hardy
Sumac, Smooth	<i>Rhus glabra</i>	10-12'	12'	Hardy
Sumac, Staghorn	<i>Rhus typhina</i>	20'	15-20'	Hardy
Tamarisk, Common	<i>Tamarix hispida</i>	10-12'	8-10'	Hardy
Tamarisk, African	<i>Tamarix africana</i>	10-12'	10'	SE
Viburnum, Arrowwood	<i>Viburnum dentatum</i>	8-10'	10-12'	NC, P
Viburnum, Nannyberry	<i>Viburnum lentago</i>	8-12'	10-12'	Hardy
Viburnum, Siebold	<i>Viburnum sieboldi</i>	12-15'	10-12'	NC
Viburnum, Tea	<i>Viburnum setigerum</i>	10'	8-10'	NC
Wahoo	<i>Euonymus atropurpureus</i>	10-15'	15-18'	Hardy
Wayfaring Tree	<i>Viburnum lantana</i>	10-12'	12'	NC
Willow, Corkscrew	<i>Salix matsudana</i> 'tortuosa'	12-15'	12'	X, P
Willow, Pussy	<i>Salix discolor</i>	10-12'	12'	X, P

DECIDUOUS SHRUBS: Medium (between 7 and 10 feet in height)

Barberry, Mentor	<i>Berberis mentorensis</i>	6-8'	5-7'	Hardy
Barberry, Truehedge	<i>Berberis thunbergi</i> 'erecta'	6-8'	4-6'	X, P
Bayberry, Northern	<i>Myrica</i> <i>pennsylvanica</i>	6-8'	4-6'	Hardy
Beautybush	<i>Kolkwitzia amabilis</i>	7-10'	8-10'	Hardy
Burning Bush, Winged	<i>Euonymus alatus</i>	8-10'	8-10'	X, P
Butterfly Bush	<i>Buddleia davidi</i>	6-8'	6'	NC
Cherry, Nanking	<i>Prunus tomentosa</i> 'Orient'	9'	10-12'	Hardy
Cotoneaster, Large Flowering	<i>Cotoneaster</i> <i>multiflora</i>	8'	8-12'	Hardy
Crabapple, Sargent	<i>Malus sargentii</i>	6-8'	8-10'	Hardy
Deutzia, Lemoine	<i>Deutzia X lemoinei</i>	5-7'	5-7'	NC, P
Deutzia, 'Pride of Rochester'	<i>Deutzia scabra</i>	6-8'	4-6'	NC, P
Dogwood, Coral	<i>Cornus alba</i> 'Siberica'	9'	8-10'	NC
Dogwood, Redosier	<i>Cornus stolonifera</i>	8'	8-10'	X, P
Dogwood, Silky	<i>Cornus ammomum</i>	8-10'	8-10'	X, P
Dogwood, Yellowtwig	<i>Cornus stolonifera</i> 'flaviramea'	8'	8-10'	X, P
Forsythia, Beatrix Farrand	<i>Forsythia x</i> <i>intermedia</i>	6-8'	6-8'	X, P
Forsythia, Golden Bell	<i>Forsythia</i> <i>viridissima</i>	6-8'	5-6'	X, P
Forsythia, 'Karl Sax'	<i>Forsythia X</i> <i>intermedia</i>	6-8'	6-8'	X, P
Forsythia, 'Lynwood Gold'	<i>Forsythia X</i> <i>intermedia</i>	8-10'	8'	X, P
Forsythia, 'Showy Border'	<i>Forsythia X</i> <i>intermedia</i>	7-8'	6-8'	X, P
Forsythia, 'Siebold'	<i>Forsythia suspensa</i> 'Sieboldii'	8-10'	10-15'	Hardy

Forsythia, 'Spring Glory'	<i>Forsythia X intermedia</i>	7-8'	6-8'	X, P
Holly, Winterberry	<i>Ilex verticillata</i>	6-8'	3-5'	NC, P
Honeysuckle, Morrow	<i>Lonicera morrowi</i>	6-8'	8-10'	Hardy
Honeysuckle, Zabel	<i>Lonicera tatarica</i> 'Zabelii'	7-10'	7-10'	Hardy
Jetbead, Black	<i>Rhodotypos scandens</i>	5-7'	6-8'	NC, P
Lilac, Late	<i>Syringa villosa</i>	8-10'	8-10'	Hardy
Lilac, Persian	<i>Syringa X persica</i>	6-8'	7-9'	NC
Magnolia, Star	<i>Magnolia stellata</i>	8-12'	10-12'	SC, P
Mockorange, 'Enchantment'	<i>Philadelphus X lemoine</i>	8'	6-8'	Hardy
Mockorange, 'Innocence'	<i>Philadelphus X lemoine</i>	8'	6-8'	Hardy
Mockorange, 'Minnesota Snowflake'	<i>Philadelphus X virginialis</i>	8'	6-8'	Hardy
Mockorange, Sweet	<i>Philadelphus coronarius</i>	8-10'	6-7'	Hardy
Ninebark, Common	<i>Physocarpus opulifolius</i>	7-9'	8-10'	Hardy
Ninebark, 'Goldleaf'	<i>Physocarpus opulifolius</i> 'luteus'	8-10'	8-10'	Hardy
Privet, Border	<i>Ligustrum obtusifolium</i>	8-10'	9'	X, P
Privet, Regel	<i>Ligustrum obtusifolium</i> 'regelium'	5-7'	6'	Hardy
Sand Cherry, Purpleleaf	<i>Prunus X cistena</i>	7-9'	8-10'	Hardy
Scotch Broom	<i>Cytisus scoparius</i>	6-8'	6-8'	SC
Spirea, Bridalwreath	<i>Spirea prunifolia</i> 'plena'	7-9'	8-10'	NC, P

Spirea, Vanhoutte	<i>Spirea X vanhouttei</i>	6-8'	8-10'	Hardy
Tamarisk, 'Summer Glow'	<i>Tamarix hispida</i>	8'	10-12'	Hardy
Viburnum, Doublefile	<i>Viburnum plicatum</i> ' <i>tomentosum</i> '	8-10'	8-10'	NC, P
Viburnum, Korean Spice	<i>Viburnum carlesii</i>	6-8'	6-8'	Hardy
Viburnum, Linden	<i>Viburnum dilatatum</i>	8-10'	8-12'	NC, P
Weigela, Old Fashioned	<i>Weigela florida</i>	6-8'	8-10'	X, P

DECIDUOUS SHRUBS: Small (between 4 and 6 feet in height)

Abelia, Glossy	<i>Abelia X grandiflora</i>	3-5'	3-5'	SC, P
Almond, Flowering (Cherry)	<i>Prunus glandulosa</i>	4-5'	3-4'	Hardy
Barberry, Japanese (Green)	<i>Berberis thunbergii</i>	4-6'	3-5'	X, P
Barberry, Redleaf	<i>Berberis thunbergii</i> ' <i>atropurpurea</i> '	4-5'	3-5'	X, P
Barberry, Kobold	<i>Berberis thunbergii</i> ' <i>Kobold</i> '	4-6'	5-6'	X, P
Beautyberry, Japanese	<i>Callicarpa japonica</i>	4-5'	4-5'	NC, P
Burning Bush, Dwarf Winged	<i>Euonymus alatus</i> ' <i>compacta</i> '	4-5'	4-5'	X, P
Cotoneaster, Spreading	<i>Cotoneaster</i> <i>divaricata</i>	5-6'	6-8'	X, P
Cranberry, Compact European	<i>Viburnum opulus</i> ' <i>compactum</i> '	4-6'	4-6'	Hardy
Honeysuckle, Clavey's Dwarf	<i>Lonicera xylosteum</i>	4-6'	4-6'	Hardy
Honeysuckle, Winter (Fragrant)	<i>Lonicera</i> <i>fragrantissima</i>	5-7'	6'	Hardy
Hydrangea, A. G.	<i>Hydrangea</i> <i>arborescens</i> ' <i>grandiflora</i> '	3-5'	5-7'	NC, P

Hydrangea, 'Nikko Blue'	<i>Hydrangea macrophylla</i>	4-5'	5-6'	X, P
Indian-current Coralberry	<i>Symphoricarpos orbiculatus</i>	3-5'	4-5'	Hardy
Kerria	<i>Kerria japonica</i>	4-6'	5-7'	X, P
Lilac, Dwarf Korean	<i>Syringa oblata 'palibiniana'</i>	4-6'	6-8'	NC, P
Mockorange, 'Albatre'	<i>Philadelphus X virginalis</i>	4-6'	4-6'	Hardy
Mockorange, 'Belle Etoile'	<i>Philadelphus X lemoine</i>	6'	6'	Hardy
Mockorange, 'Bouquet Blanc'	<i>Philadelphus X virginalis</i>	6'	6'	Hardy
Mockorange, 'Glacier'	<i>Philadelphus X virginalis</i>	4-6'	4-6'	Hardy
Quince, Japanese Flowering	<i>Chaenomeles lagenaria (speciosa)</i>	5-7'	6-8'	X, P
Snowberry	<i>Symphoricarpos albus</i>	4-6'	4-6'	Hardy
Spirea, Garland	<i>Spirea X arguta</i>	5-6'	5-6'	X, P
Spirea, Thunberg	<i>Spirea thunbergi</i>	4-5'	4-5'	Hardy
Viburnum, Wright's	<i>Viburnum wrighti 'hessei'</i>	3-5'	4-6'	X, P
Weigela, 'Bristol Ruby'	<i>Weigela X florida</i>	4-6'	4-6'	X, P
Weigela, 'Eva Rathke'	<i>Weigela X floribunda</i>	4-6'	6-8'	X, P
Weigela, 'Java Red'	<i>Weigela purpurea</i>	5-7'	6-8'	X, P

DECIDUOUS SHRUBS: Dwarf (less than 4 feet in height)

Barberry, Crimson Pygmy	<i>Berberis thunbergi 'nana'</i>	1½-2'	2-3'	X, P
Chenault Coralberry	<i>Symphoricarpos X chenaulti</i>	3-4'	3-4'	Hardy
Cinquefoil	<i>Potentilla fruticosa</i>	2-4'	2-4'	Hardy

Cotoneaster, Cranberry	<i>Cotoneaster apiculata</i>	2-3'	3-4'	X, P
Deutzia, Rosepanicle	<i>Deutzia X rosea 'eximia'</i>	3-4'	4-5'	NC
Deutzia, Slender	<i>Deutzia gracilis</i>	2-3'	3-4'	NC
Forsythia, 'Bronx'	<i>Forsythia viridissima 'Bronxensis'</i>	1½-2'	3'	X, P
Hydrangea, 'Annabelle'	<i>Hydrangea X arborescens</i>	3-4'	4-5'	X, P
Mockorange, Silver Showers	<i>Philadelphus X lemoine</i>	3-4'	3-4'	Hardy
Mockorange, 'Dwarf Minnesota Snowflake'	<i>Philadelphus X virginialis 'nana'</i>	2-3'	2-3'	Hardy
Mockorange, Dwarf Sweet	<i>Philadelphus coronarius 'nana'</i>	3-4'	3-4'	Hardy
Ninebark, Dwarf	<i>Physocarpus opulifolius 'nanus'</i>	3-4'	3-4'	Hardy
Pea Shrub, Pygmy	<i>Caragana arborescens</i>	1½-2'	1½-2'	Hardy
Quince, Dwarf Flowering	<i>Chaenomeles meulei</i>	2-3'	2-3'	X, P
Quince, Japanese Flowering	<i>Chaenomeles japonica</i>	3-4'	4-5'	Hardy
Quince, Red Japanese	<i>Chaenomeles japonica 'rubra'</i>	3-4'	4-5'	Hardy
St. Johnswort, Hidcote	<i>Hypericum patulum 'Hidcote'</i>	1½-2'	1½-2'	X
St. Johnswort, Kalm	<i>Hypericum kalmianum</i>	3-4'	3-4'	Hardy
St. Johnswort, Sungold	<i>Hypericum patulum 'Sungold'</i>	1½-2'	1½-2'	Hardy
Spirea, Anthony Waterer	<i>Spirea X bumalda 'Anthony Waterer'</i>	2-3'	2-3'	X, P

Spirea, Froebel	<i>Spirea X bumalda</i> 'Froebeli'	3-4'	3-4'	X, P
Spirea, Redflowering	<i>Spirea japonica</i> 'coccinea'	3-4'	3-4'	X, P
Spirea, Snowmound	<i>Spirea nipponica</i> 'Snowmound'	3-4'	3-4'	Hardy
Willow, Dwarf Arctic Blue	<i>Salix purpurea</i> 'nana'	3-4'	3-4'	NC

15

[1]

Common Name Botanical Name Height Spread Adaptability

BROADLEAF EVERGREEN SHRUBS: Tall (10 feet or more in height)

Holly, American	<i>Ilex opaca</i>	12-15'	8-10'	SC, P
Holly, Yaupon	<i>Ilex vomitoria</i>	10-15'	8-10'	SC, P
Magnolia, Southern	<i>Magnolia grandiflora</i>	15-20'	10-12'	SC, P

BROADLEAF EVERGREEN SHRUBS: Medium (between 7 and 10 feet in height)

Euonymus, Large-Leaf	<i>Euonymus kiautschovicus</i> (<i>Patens</i>)	6-8'	8-10'	Hardy
Euonymus, 'Manhattan'	<i>Euonymus kiautschovicus</i> (<i>Patens</i>)	8-10'	8-12'	Hardy
Mahonia, Oregon Grape	<i>Mahonia aquifolium</i>	5-7'	4-5'	Hardy
Pyracantha, 'Kasan'	<i>Pyracantha coccinea</i> 'Kasan'	6-8'	6-8'	X, P
Pyracantha, 'Lalandii'	<i>Pyracantha coccinea</i> 'Lalandii'	5-10'	8-10'	X, P
Pyracantha, 'Wyatti'	<i>Pyracantha coccinea</i> 'Wyatti'	6-8'	6-8'	X, P
Viburnum, Leatherleaf	<i>Viburnum rhytidophyllum</i>	8-12'	8-12'	NC, P

BROADLEAF EVERGREEN SHRUBS: Small (between 4 and 6 feet in height)

Barberry, Wintergreen	<i>Berberis juliana</i>	4-6'	3-5'	NC, P
Euonymus, Dupont	<i>Euonymus kiautschovicus (Patens)</i>	5-6'	4-6'	Hardy
Euonymus, 'Emerald 'n' Gold'	<i>Euonymus fortunei 'vegetus'</i>	4-5'	4'	X, P
Euonymus, 'Jewel'	<i>Euonymus kiautschovicus (Patens)</i>	5-6'	4-6'	Hardy
Euonymus, Medium Leaf	<i>Euonymus kiautschovicus (Patens)</i>	5-6'	4-6'	Hardy
Euonymus, 'Sarcoxie'	<i>Euonymus fortunei 'vegetus'</i>	4-5'	4'	X, P
Holly, Dwarf Japanese	<i>Ilex crenata 'compacta'</i>	4-5'	4-5'	SC, P
Holly, 'Hellers' Japanese	<i>Ilex crenata 'Helleri'</i>	4'	5'	SC, P
Holly, Hetz Japanese	<i>Ilex crenata 'Hetzii'</i>	4'	5'	SC, P
Holly, Dwarf Yaupon	<i>Ilex vomitoria 'nana'</i>	4'	5'	SC, P
Nandina	<i>Nandina domestica</i>	4-5'	4'	SC, P
Pyracantha, 'Low Boy'	<i>Pyracantha coccinea 'Low Boy'</i>	4-5'	4-5'	X, P

BROADLEAF EVERGREEN SHRUBS: Dwarf (less than 4 feet in height)

Azalea, Kurume	<i>Rhododendron obtusum</i>	3'	2-3'	SC, P
Azalea, Mollis Hybrid	<i>Rhododendron X kosterianum</i>	3-4'	3-4'	SC, P

Boxwood, Edging	<i>Buxus sempervirens</i>	2-3'	2-3'	NC, P
Boxwood, Wintergreen	<i>Buxus microphyllus</i>	4'	5'	NC, P
Euonymus, 'Emerald Cushion'	<i>Euonymus fortunei</i> 'vegetus'	3-4'	3-4'	X, P
Euonymus, 'Emerald Gaiety'	<i>Euonymus fortunei</i> 'vegetus'	3-4'	2-3'	X, P
Euonymus, Small Leaf	<i>Euonymus</i> <i>kiautschovicus</i> (<i>Patens</i>)	2-3'	2-3'	Hardy
Euonymus, Vinca Leaf	<i>Euonymus</i> <i>kiautschovicus</i> 'vincafolia'	3-4'	3-4'	X, P
Holly, Dwarf Burford	<i>Ilex cornuta</i> 'Burfordi <i>nana</i> '	3-4'	2-3'	Hardy
Rhododendron	<i>Rhododendron</i> <i>catawbiense</i>	4'	4'	SC, P

[1]

Adaptability:

SE—hardy in southeastern Kansas only;

SC—hardy in south central and eastern Kansas;

NC—hardy in central and eastern Kansas only;

P—protect from south west winds;

X—hardiness is doubtful in hotter, drier locations of Kansas;

Hardy—fully hardy in exposed areas of Kansas.

PROBLEMS TO EXPECT

Every shrub species grown in Kansas can be expected to be afflicted with some problem during its lifetime. The most common problems are insect and disease attacks. These usually can be controlled by routine spraying. Knowledge of the problems make it easier to diagnose and correct before they become severe.

Soil alkalinity or high pH may also limit the use of some plants. Iron chlorosis is a common problem with some shrubs grown in the western half of Kansas. This is easily corrected by applying iron sulfate or iron chelate to the plant or soil.

Your local County Extension Agent can assist you with the diagnosis of any disease, insect, or cultural problem.



All Kansas Extension educational programs and materials are available to all individuals without discrimination on the basis of race, color, national origin, sex, or religion.

**Cooperative Extension Service, Kansas State University,
Manhattan**

C-469

February 1973

Issued in furtherance of Cooperative Extension work, acts of May 8 and June 30, 1914, in cooperation with U. S. Department of Agriculture. Robert A. Bohannon, Director of Extension, Kansas State University of Agriculture and Applied Science.

34-6672

2-73—20M

KANSAS STATE PRINTING PLANT

Transcriber's Notes

- Silently corrected a few typos.
- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- In the text versions only, text in *italics* is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK SHRUBS FOR
KANSAS LANDSCAPES ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.